Native Sparse Attention for Optical Remote Sensing Spectral Images

Xuyang Li, Chenyu Li, Lei Wang, *Member, IEEE*, Keming Chen, *Member, IEEE*, and Danfeng Hong, *Senior Member, IEEE*

***Abstract*—Optical remote sensing (RS) spectral images contain rich cross-band response patterns. Recent RS foundation models (RSFMs) improve compatibility with variable-band inputs through unified or channelwise tokenization, but such input-interface compatibility does not ensure generalization to unseen sensors. Moreover, existing backbones often apply dense self-attention to flattened spatial–spectral tokens, resulting in high floating point operations per second (FLOPs) and memory costs while mixing heterogeneous interactions indiscriminately. To address this issue, we propose spectral-native sparse attention (SpectralNSA), a spectral-native sparse attention (NSA) backbone for optical RS spectral images. SpectralNSA models cross-channel tokens at each spatial location using three complementary sparse patterns: blockwise spectral compression for global context, group-shared Top-k block selection for informative spectral regions, and local spectral windows for neighboring-band continuity. We further introduce a spectral-shared-spatial (SSS) structure that compresses channelwise features into shared-spatial prototypes, performs compact spatial modeling, and reinjects spatial context through gated residual connections, thereby reducing redundant per-channel–spatial interactions. Extensive experiments on scene classification, urban and agricultural segmentation, hyperspectral driving-scene segmentation, and road extraction demonstrate consistent accuracy gains and favorable scalability, especially for moderate- and high-dimensional spectral inputs. A within-sensor variable-band evaluation further shows that a model trained with the complete channel set can directly process shortened wavelength-ordered sequences after unavailable bands are removed, without retraining. At a fixed spatial resolution, the FLOPs advantage of SpectralNSA over ChannelViT increases with the number of spectral channels, reaching 140.94× for the Tiny model with 150 input channels. Controlled pretraining and transfer experiments further demonstrate its effectiveness for transferable spectral–spatial representation learning across classification and dense prediction tasks.**

***Index Terms*—Optical, remote sensing (RS), vision Transformer.**

I. Introduction

In RECENT years, Transformers and foundation models have advanced rapidly in remote sensing (RS) and have been widely applied to land-cover classification, change detection, and multimodal understanding [1], [2], [3]. Unlike natural images with fixed RGB inputs, optical RS data may involve varying numbers of spectral bands, substantial inter-band redundancy, and complex cross-sensor heterogeneity [4]. Consequently, enabling Transformers to handle variable-band spectral inputs has become an important problem in RS foundation models (RSFMs). Existing studies improve input compatibility through shared patch embedding, channel mapping, band reordering, wavelength encoding, or unified tokenization [5], [6]. Here, we distinguish such variable-band input compatibility from unrestricted cross-sensor generalization. Fig. 1 conceptually illustrates representative cross-band response differences that motivate structured spectral modeling.

Input compatibility, however, does not necessarily imply backbone suitability. After spectral inputs are converted into tokens, many RSFMs still rely on dense self-attention over a flattened sequence. This mismatch is especially evident under channelwise tokenization, which preserves band-specific information by splitting each channel or channel group into spatial patches [7], [8]. Although this avoids premature spectral mixing, the sequence length grows approximately with spatial patches × spectral channels. Dense self-attention over this expanded spatial–spectral sequence, therefore, increases floating point operations per second (FLOPs) and memory rapidly because attention scales quadratically with sequence length [9], [10]. This creates a structural mismatch: the input interface is adapted to variable spectral configurations, whereas the backbone still treats the resulting tokens as a generic flattened sequence. The key question is therefore how to design a scalable backbone that remains aligned with wavelength-ordered spectral imagery after tokenization.

The resulting cost also reflects a modeling issue. Dense attention mixes same-location cross-channel relations, same-channel cross-location relations, and arbitrary cross-location cross-channel relations within one unconstrained attention map. Given the spectral redundancy and limited labeled data common in RS [11], many of these interactions may be

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Xuyang Li, Lei Wang, and Keming Chen are with the Key Laboratory of Target Cognition and Application Technology (TCAT), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: lixuyang23@mails.ucas.ac.cn; wanglei002931@aircas.ac.cn; kmchen.ie@gmail.com).

Chenyu Li is with INRAE, Université Grenoble Alpes, 38000 Grenoble, France (e-mail: chenyuli.erh@gmail.com).

Danfeng Hong is with School of Automation, Southeast University, Nanjing 210096, China (e-mail: danfeng.hong@seu.edu.cn).

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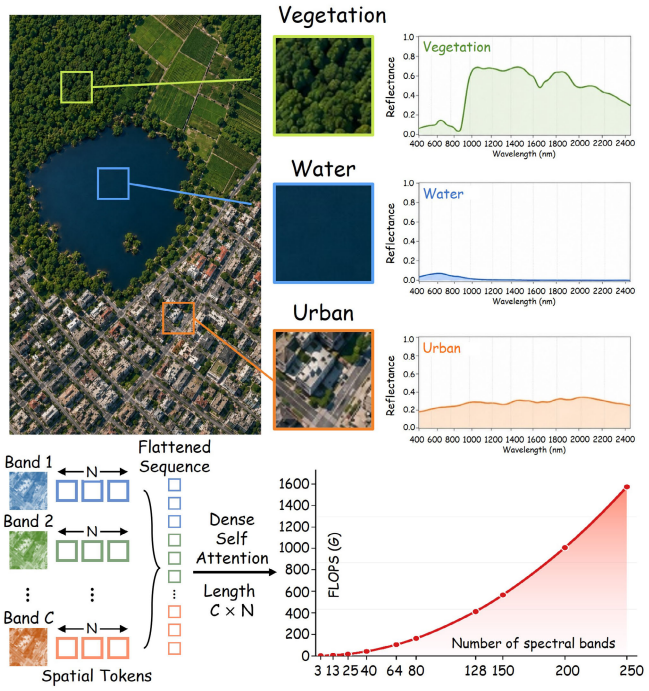


Fig. 1. Conceptual motivation of spectral-NSA. Channelwise tokenization preserves band-specific spectral information but expands the token sequence from spatial tokens to spatial–spectral tokens. Since dense self-attention constructs pairwise interactions over the flattened sequence, its computational and memory costs grow rapidly as the number of spectral bands increases. This motivates a backbone that models spectral dependencies with structured sparsity rather than dense global attention. The displayed spectral profiles are illustrative examples rather than a complete statistical analysis of all dataset classes.

low value. On the contrary, discriminative information in optical spectral imagery often depends on cross-band response patterns at the same spatial location [4], [12]. Neighboring bands exhibit continuity, distant spectral regions may provide complementary cues, and only part of the spectrum may be informative for a given target. Thus, an effective backbone should organize sparse attention according to spectral structure rather than merely reduce the number of interactions.

Existing efficient attention methods address parts of this problem. Spatial window attention exploits image-plane locality [13], while band selection and pruning reduce spectral redundancy [14]. Generic sparse or approximate attention also lowers computation [15]. However, these strategies are not specifically organized around wavelength-ordered spectral continuity, local-and-long-range cross-band complementarity, and regular blockwise execution. The challenge is therefore not simply to make attention sparse, but to make the sparsity spectral native.

Motivated by these observations, we propose spectral native sparse attention (SpectralNSA), a spectral-native sparse attention (NSA) backbone for optical spectral RS images. SpectralNSA follows the general compress–select–window principle of NSA but redesigns it around the wavelength-ordered sequence of cross-channel tokens at each spatial location instead of directly applying the language-sequence formulation to flattened spatial–spectral tokens. Blockwise spectral compression captures coarse global spectral context, group-shared Top-k routing focuses computation on informative spectral regions, and local spectral windows preserve neighboring-band continuity. These branches correspond to complementary properties of spectral imagery: global summarization, task-adaptive spectral relevance, and local smoothness. By integrating them within one blockwise sparse mechanism, SpectralNSA makes cross-band dependency modeling a primary backbone operation rather than an auxiliary module attached to dense attention.

The key contribution of SpectralNSA lies in redefining the sparse coordinate system and backbone information flow for optical spectral imagery. Sparse access is organized over same-location, wavelength-ordered sequences; contiguous blocks and local windows correspond to spectral regions and neighborhoods; and spatial context is introduced after spectral modeling through the proposed shared-spatial pathway. This spectral-native organization forms the core of the proposed backbone design.

In particular, we further introduce a spectral-shared-spatial (SSS) organization to reduce redundant per-channel–spatial modeling. SSS first performs spectral-native sparse modeling, then distills channelwise features into a small set of shared-spatial prototypes through learnable fusion. Compact spatial attention is applied to these prototypes, and the resulting context is reinjected into channelwise representations through gated residual connections. This spectral-first/shared-spatial flow preserves spatial complementarity without repeatedly applying spatial attention to every channel.

SpectralNSA is additionally organized as regular blockwise sparse computation. Contiguous spectral blocks provide a shared sparse coordinate system, group-shared routing reduces fragmented per-head access, and spatial chunked execution controls intermediate activation memory without changing the per-location spectral result. Extensive experiments across scene classification, urban and agricultural segmentation, hyperspectral driving-scene segmentation, road extraction, efficiency analysis, and controlled pretraining/transfer settings validate the proposed design. Results across model scales and channel numbers show favorable scaling behavior, particularly for moderate- and high-dimensional spectral inputs. The main contributions are summarized as follows.

1) We propose **SpectralNSA**, which reorganizes the established compress–select–window sparse attention principle along wavelength-ordered, same-location spectral sequences using contiguous spectral blocks, group-shared Top-k routing, and local spectral neighborhoods.
2) We introduce the **SSS** backbone architecture, which performs spectral-first modeling, constructs shared-spatial prototypes, applies compact spatial attention, and reinjects the resulting context through gated residual connections.
3) We organize SpectralNSA with regular blockwise sparse access and memory-aware spatial chunking, reducing fragmented sparse indexing and controlling activation memory.
4) Extensive experiments demonstrate a favorable accuracy–scalability tradeoff, verify scaling across model sizes and channel numbers, and demonstrate

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the effectiveness of SpectralNSA as a competitive pretrainable backbone across diverse optical spectral settings.

II. Related Work

A. Transformers for Variable-Band Spectral Inputs

Transformers and foundation models have extended RS representation learning from fixed RGB imagery to multispectral, hyperspectral, SAR, and multimodal Earth observation. Early work adapted the ViT/masked-autoencoding (MAE) paradigm to geospatial data: RVSA [3] improves spatial modeling with rotated, varied-size window attention, S2MAE [16] introduces spectral positional encodings for satellite imagery, and scale-MAE [17] incorporates scale-aware positional encoding and multiscale reconstruction.

More recent studies focus on heterogeneous or variable-band spectral inputs. ChannelViT [7] patchifies each channel independently and preserves channel identity with learnable embeddings. SpectralGPT [2] develops a unified generative spectral pretraining framework. DOFA [18] represents bands by wavelengths and uses a hypernetwork to generate model weights from central wavelengths, while HyperFree [5] constructs variable-channel embeddings from a learnable dictionary spanning the spectral range. SpecAware [19] further combines sensor meta-attributes and semantic features for cross-sensor hyperspectral mapping. Together, these approaches show that explicit channel or wavelength information is useful for handling heterogeneous spectral configurations. These methods improve variable-band and, in some cases, cross-sensor adaptability, but largely focus on the input interface. On the contrary, SpectralNSA focuses on backbone computation after channelwise tokenization, where channel-expanded spatial–spectral sequences make dense attention increasingly expensive and redundant.

B. Sparse Attention for Efficient Long-Sequence Modeling

The quadratic cost of self-attention has motivated local-window, block-sparse, hashing-based, and linear/approximate attention [20]. Longformer [21] combines sliding windows with global tokens, BigBird [22] uses local, random, and global block-sparse patterns, Reformer [23] employs locality-sensitive hashing, Performer [24] approximates softmax attention with random features, and Swin transformer [13] restricts attention to shifted spatial windows. Recent work moves toward dynamic and trainable sparsity. NSA [25] combines compression, selection, and local attention as an end-to-end trainable mechanism rather than a posthoc approximation of dense attention. VideoNSA [26] extends NSA to long video sequences, and block-sparse FlashAttention [27] emphasizes the execution benefits of regular blockwise sparsity. These methods are designed mainly for text, video, or generic spatial tokens. SpectralNSA instead reorganizes the compress–select–window mechanism along the optical spectral axis at each spatial location and couples it with shared-spatial prototypes and gated context reinjection.

C. Spectral–Spatial Modeling for Spectral Images

Joint spectral–spatial representation is a long-standing problem in spectral image analysis. HybridSN [28] combines 3-D and 2-D convolutions to capture joint spectral–spatial features. Transformer-based approaches increasingly formulate spectral modeling as sequence learning: SpectralFormer [29] uses groupwise spectral embedding and cross-layer fusion; SSFTT [30] combines convolutional features with transformer tokenization; HiT [31] embeds convolution into transformer blocks; SST [32] integrates spatial and spectral modeling; and MambaX [33] studies state fusion in the spectral domain. These methods demonstrate the importance of joint spectral–spatial learning, but many rely on early fusion, parallel fusion, or repeated spatial processing across channels. On the contrary, SpectralNSA first reorganizes same-location cross-band dependencies along the spectral axis and then introduces spatial context through a compact shared-prototype pathway, yielding a spectral-first information flow. Large-scale visual pretraining and analysis-ready embeddings provide a complementary direction. DINO-based representations [34], AlphaEarth [35], TESSERA [36], and related products [37] primarily target transferable semantic pretraining or reusable geospatial embeddings. SpectralNSA instead addresses the backbone used to process long channelwise optical spectral sequences; RGB-centric checkpoints generally require spectral adaptation, modified patch embedding, or additional pretraining before accepting tens or hundreds of bands.

III. Method

A. Design Principles

Our method is motivated by the observation that a unified input interface for variable-band optical spectral data does not by itself define a suitable transformer backbone. Channelwise tokenization preserves band identity and supports varying channel configurations, but it also exposes a structured token space with two different dependency axes: spectral interactions among bands at the same spatial location and spatial interactions across patches. Directly applying dense attention to the flattened channel–spatial sequence mixes these two dependencies and introduces unnecessary cost. Therefore, we adopt a spectral-first design, where cross-band tokens at each spatial location are first treated as a spectral sequence and modeled before spatial context is introduced.

Let the tokenized input be $\mathbf{X} \in \mathbb{R}^{C\times N\times D}$, where C, N, and D denote the number of channels, spatial patches, and embedding dimension, respectively. For each spatial location n, $\mathbf{X}_{:,n,:} \in \mathbb{R}^{C\times D}$ forms a spectral sequence. SpectralNSA models this sequence with structured sparse attention based on contiguous spectral blocks, combining blockwise compression, group-shared Top-k block selection, and local spectral windows. After spectral dependencies have been reorganized, spatial information is introduced through a shared-spatial design, where multichannel features are distilled into compact spatial prototypes and then injected back into channelwise representations. As summarized in Fig. 2, the overall backbone follows a spectral-first and shared-spatial principle: spectral

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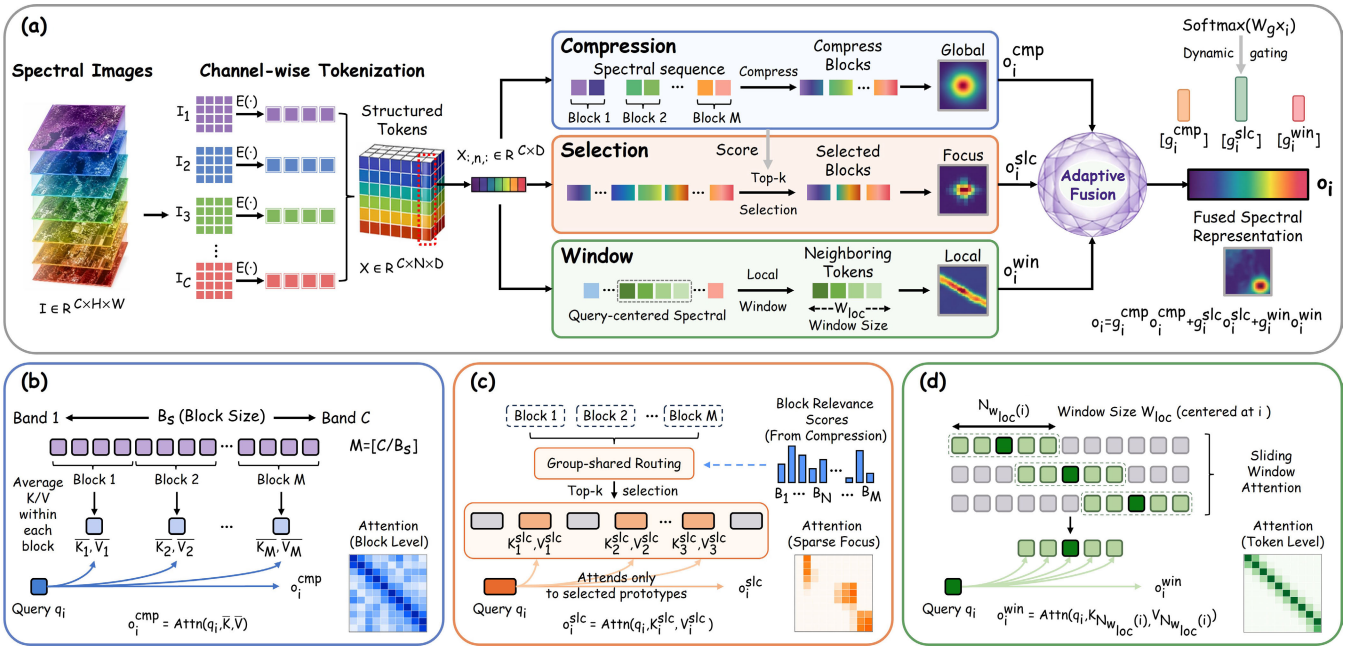


Fig. 2. Overview of SpectralNSA. The input spectral image is first converted into structured channelwise tokens. SpectralNSA then models cross-band dependencies at each spatial location through a compress–select–window sparse attention mechanism, followed by shared-spatial modeling and gated context reinjection. (a) SpectralNSA. (b) Blockwise compression. (c) Group-shared Top-k block selection. (d) Local spectral window.

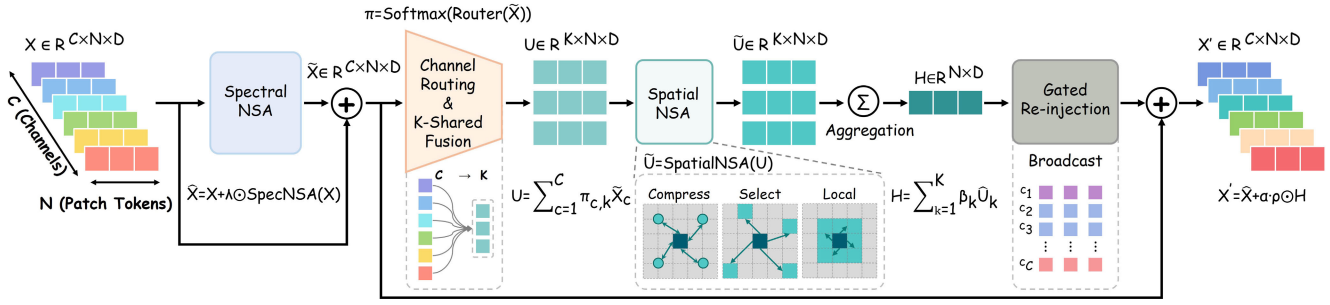


Fig. 3. Overview of the SSS block. The block first applies spectral-NSA at each spatial location to reorganize cross-band dependencies. The updated channelwise features are then routed into a small set of shared-spatial prototypes, on which compact sparse spatial attention is performed. Finally, the aggregated spatial context is broadcast back to all channels through gated residual reinjection, reducing redundant per-channel–spatial modeling while preserving spatial complementarity.

dependencies are explicitly prioritized, while spatial modeling is performed in a compact and low-redundancy manner.

B. Structured Channelwise Tokenization

Let the input RS image be $\mathbf{I} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, image height, and image width, respectively. To avoid prematurely mixing different bands at the input stage, we apply patch embedding independently to each channel. Let $\mathcal{E}(\cdot)$ denote a shared single-channel patch tokenizer that maps each channel image into $N = H_p W_p$ patch tokens. The token corresponding to channel c and patch n is defined as

$$\mathbf{z}_{c,n} = \mathcal{E}\left(\mathbf{I}_c\right)_n + \mathbf{p}_n + \mathbf{e}_c \tag{1}$$

where $\mathbf{p}_n \in \mathbb{R}^D$ is a shared-spatial positional embedding and $\mathbf{e}_c \in \mathbb{R}^D$ is a channel identity embedding. Stacking all tokens yields

$$\mathbf{X}_{c,n,:} = \mathbf{z}_{c,n}, \quad \mathbf{X} \in \mathbb{R}^{C\times N\times D}. \tag{2}$$

For each sensor configuration, optical bands are arranged in ascending order of center wavelength before tokenization. The embedding $\mathbf{e}_c$ is indexed by the canonical identity of the band in the complete wavelength-ordered sensor configuration.

For a shortened variable-band input, let $\mathcal{S} = \{s_1, \ldots, s_{C'}\}$ denote the retained canonical band indices. Unavailable bands are physically removed, and the retained tokens continue to use $\mathbf{e}_{s_1}, \ldots, \mathbf{e}_{s_{C'}}$ rather than being reassigned new identities from 1 to C'. Padding is introduced only for block-size alignment, and padded positions are excluded from attention.

The default implementation uses canonical channel-index embeddings; we additionally evaluate a center-wavelength descriptor produced by a shared two-layer MLP while keeping the tokenizer and backbone unchanged.

This representation preserves the two-axis *channel* × *spatial position* structure rather than flattening all tokens into a

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TABLE I

COMPLETE STAGEWISE CONFIGURATION OF SPECTRALNSA. B_s, K_s, AND W_s DENOTE THE SPECTRAL BLOCK SIZE, SELECTED TOP-k BLOCK NUMBER, AND LOCAL SPECTRAL WINDOW SIZE; B_p, K_p, AND W_p ARE THE CORRESPONDING SHARED-SPATIAL SETTINGS. G_1/G_2 GIVES THE ROUTING-GROUP COUNTS IN STAGES 1/2, AND K_{proto} IS THE NUMBER OF SHARED-SPATIAL PROTOTYPES. SPARSE BUDGETS LISTED ONCE PER SCALE ARE SHARED BY BOTH STAGES UNLESS OTHERWISE SPECIFIED

Scale	Patch	Embed dim.	Depths	Heads	B_s	K_s	W_s	B_p	K_p	W_p	G_1/G_2	K_{proto}
Tiny	4	64/128	2/2	2/4	2	1	3	8	2	3	1/2	2
Small	4	192/384	4/4	3/6	2	2	5	8	2	5	1/3	2
Base	8	384/768	3/6	6/12	2	2	5	8	2	5	3/6	4
Large	8	512/1024	6/12	8/16	2	2	5	8	2	5	4/8	4

TABLE II

QUANTITATIVE COMPARISON ON **EuroSAT**. WE REPORT TOP-1 ACCURACY (%) UNDER FAMILY SPECIFIC TINY/SMALL/BASE CAPACITY TIERS. FOR THE RESNET BASELINE, THESE THREE TIERS CORRESPOND TO RESNET-18, RESNET-50, AND RESNET-101, RESPECTIVELY. THESE TIERS ARE NOT STRICT ISO-PARAMETER OR ISO-FLOPS MATCHES

Method	Tiny	Small	Base
ResNet	94.85	95.98	96.55
ViT	93.44	93.85	94.09
VMamba	94.59	94.96	95.41
ChannelViT	91.24	94.63	95.92
Swin	95.41	95.57	95.86
SpectralNSA	**95.88**	**96.89**	**97.04**

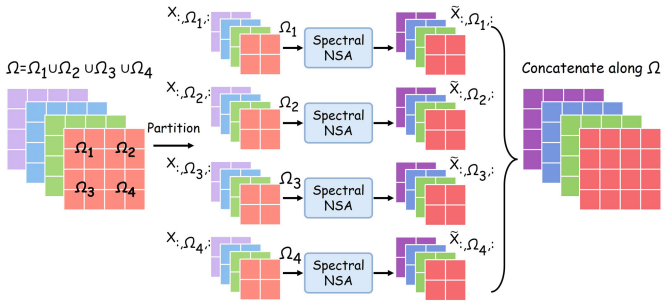


Fig. 4. Spectral chunked execution. SpectralNSA is applied independently to the spectral sequence at each spatial location. For memory control, spatial locations are partitioned into nonoverlapping chunks and processed sequentially with the same spectral attention operator. This execution strategy reduces intermediate activation size while preserving the per-location spectral attention result.

sequence of length CN, enabling spectral modeling at fixed locations and spatial modeling over shared prototypes.

C. Spectral NSA

1) Unified Multihead Formulation: Given the structured token tensor $\mathbf{X} \in \mathbb{R}^{C\times N\times D}$, the token set at a fixed spatial location n forms a spectral sequence $\mathbf{X}^{(n)} = \mathbf{X}_{:,n,:} \in \mathbb{R}^{C\times D}$. We describe the operation for this sequence and explicitly write the multihead formulation. For the hth attention head, the query, key, and value are computed as

$$\mathbf{Q}^h = \mathbf{X}^{(n)}\mathbf{W}_Q^h, \quad \mathbf{K}^h = \mathbf{X}^{(n)}\mathbf{W}_K^h, \quad \mathbf{V}^h = \mathbf{X}^{(n)}\mathbf{W}_V^h \tag{3}$$

where $\mathbf{Q}^h, \mathbf{K}^h, \mathbf{V}^h \in \mathbb{R}^{C\times d_h}$, $d_h = D/H$, and H is the number of attention heads. We denote the ith query, key, and value in head h by $\mathbf{q}_i^h$, $\mathbf{k}_i^h$, and $\mathbf{v}_i^h$, respectively.

TABLE III

QUANTITATIVE COMPARISON ON **C2Seg**

Method	P	R	F1	OA	mIoU
Tiny					
ResNet	56.77	47.19	49.29	70.50	36.95
ViT	61.99	50.17	53.32	71.45	40.26
VMamba	61.66	47.92	51.04	71.28	38.33
ChannelViT	64.42	54.44	57.18	73.03	43.97
Swin	59.39	48.75	51.19	71.13	38.56
SpectralNSA	**68.92**	**56.86**	**60.67**	**74.88**	**47.22**
Small					
ResNet	61.74	52.27	54.34	73.37	41.76
ViT	64.56	57.00	58.78	75.69	46.11
VMamba	65.80	59.44	60.83	76.82	48.23
ChannelViT	65.10	54.84	58.03	73.45	44.60
Swin	62.71	56.55	57.99	75.55	45.35
SpectralNSA	**73.04**	**65.60**	**68.25**	**79.95**	**56.13**
Base					
ResNet	64.04	52.89	55.41	73.50	42.86
ViT	69.84	59.72	63.11	76.71	50.02
VMamba	**72.50**	64.42	66.90	79.08	54.81
ChannelViT	66.93	64.46	64.67	78.46	52.26
Swin	72.33	65.28	67.95	79.09	55.68
SpectralNSA	71.83	**70.92**	**68.39**	**79.80**	**56.42**

If dense self-attention is directly used on $(\mathbf{Q}^h, \mathbf{K}^h, \mathbf{V}^h)$, every band token interacts with all other band tokens. This incurs $\mathcal{O}(C^2)$ spectral attention complexity for each spatial location and makes it difficult to highlight truly informative spectral regions. To address this, we construct a structured sparse visible set for each query through three complementary branches. Let the outputs of the three branches in head h be denoted by $\mathbf{o}_i^{\text{cmp},h}$, $\mathbf{o}_i^{\text{slc},h}$, and $\mathbf{o}_i^{\text{win},h}$, respectively. The output of the ith spectral token in head h is defined as

$$\mathbf{o}_i^h = g_i^{\text{cmp},h}\mathbf{o}_i^{\text{cmp},h} + g_i^{\text{slc},h}\mathbf{o}_i^{\text{slc},h} + g_i^{\text{win},h}\mathbf{o}_i^{\text{win},h} \tag{4}$$

where the branch weights are dynamically predicted from the input token

$$\left[g_i^{\text{cmp},h}, g_i^{\text{slc},h}, g_i^{\text{win},h}\right] = \text{Softmax}\left(\mathbf{W}_g^h\mathbf{x}_i\right). \tag{5}$$

Here, $\mathbf{x}_i = \mathbf{X}_{i,:}^{(n)}$ denotes the input spectral token. The final multihead output is obtained by concatenating all heads and applying the output projection in the standard way. The three branches provide global summarization, informative-block

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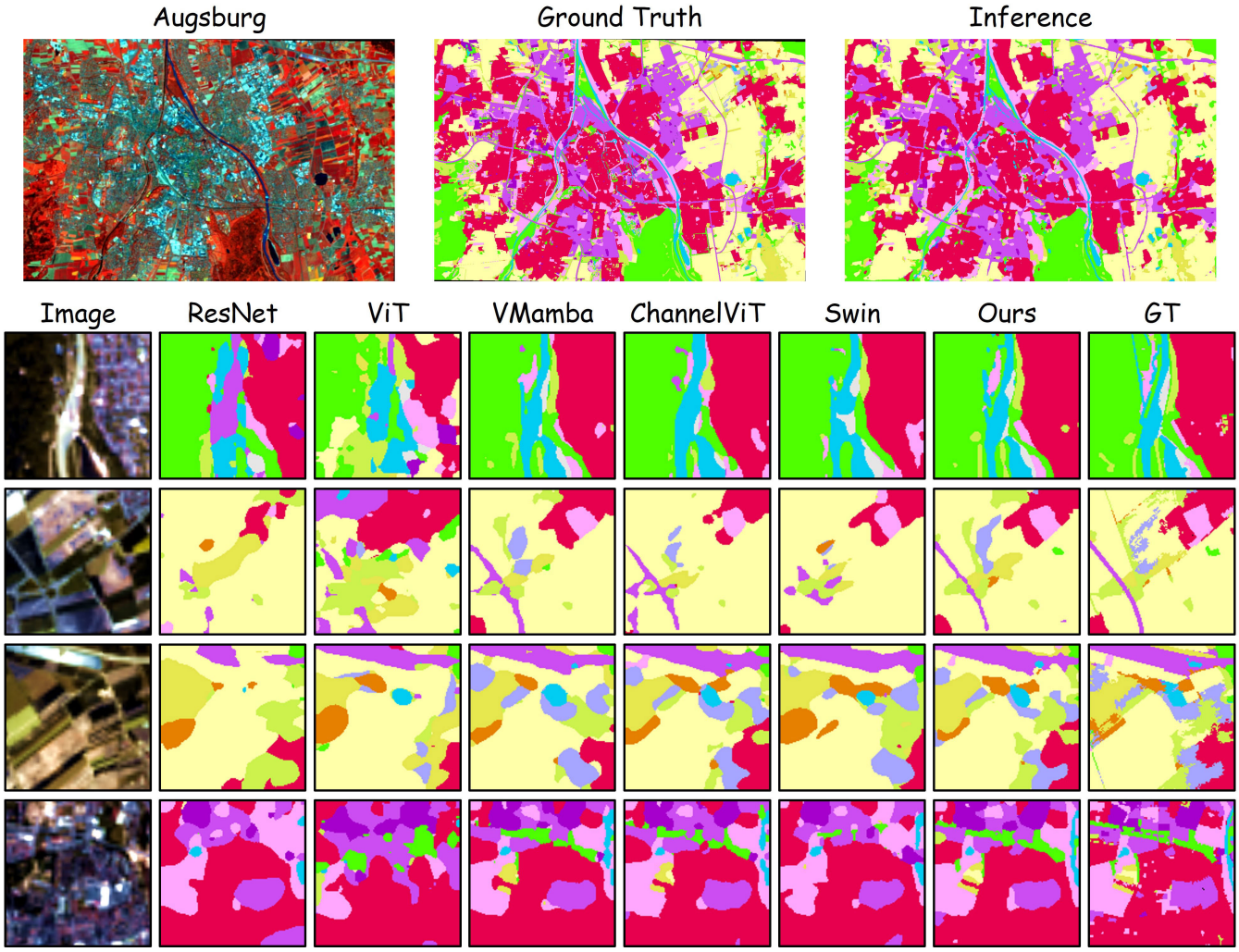


Fig. 5. Qualitative comparison on the C2Seg Augsburg region. SpectralNSA produces more coherent land-cover regions and better preserves object boundaries than the baselines.

selection, and local spectral continuity, and are adaptively fused by dynamic gating.

2) Blockwise Compression: The goal of the compression branch is to establish a coarse global understanding of the full spectral sequence without retaining all pairwise band interactions. To this end, we partition the spectral sequence along the channel axis into M contiguous spectral blocks

$$\mathcal{B}_m = \{j \mid (m-1)\,B_s < j \le \min(mB_s, C)\} \quad m = 1, \ldots, M \tag{6}$$

where B_s denotes the spectral block size and $M = \lceil C/B_s \rceil$. In implementation, the sequence can be padded to a multiple of B_s, and the padded positions are excluded by an attention mask.

For each spectral block and each head, we construct a block-level prototype by averaging the keys and values within the block

$$\bar{\mathbf{k}}_m^h = \frac{1}{|\mathcal{B}_m|} \sum_{j \in \mathcal{B}_m} \mathbf{k}_j^h, \quad \bar{\mathbf{v}}_m^h = \frac{1}{|\mathcal{B}_m|} \sum_{j \in \mathcal{B}_m} \mathbf{v}_j^h. \tag{7}$$

This yields the compressed block-level key/value sets

$$\bar{\mathbf{K}}^h = \left[\bar{\mathbf{k}}_1^h, \ldots, \bar{\mathbf{k}}_M^h\right]^\top, \quad \bar{\mathbf{V}}^h = \left[\bar{\mathbf{v}}_1^h, \ldots, \bar{\mathbf{v}}_M^h\right]^\top \tag{8}$$

where $\bar{\mathbf{K}}^h, \bar{\mathbf{V}}^h \in \mathbb{R}^{M \times d_h}$. The compression branch output is then

$$\mathbf{o}_i^{\mathrm{cmp},h} = \mathrm{Attn}\left(\mathbf{q}_i^h, \bar{\mathbf{K}}^h, \bar{\mathbf{V}}^h\right) \tag{9}$$

where

$$\mathrm{Attn}(\mathbf{q}, \mathbf{K}, \mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{d_h}}\right)\mathbf{V}. \tag{10}$$

This branch provides coarse global spectral summarization, and its block-level relevance scores are reused by the subsequent selection branch.

3) Group-Shared Top-k Block Selection: Once a coarse global view is available, the next natural question is whether attention can be further concentrated on a small number of informative spectral regions. This is the role of the selection branch. Rather than introducing an additional scoring network, we directly reuse the block-level relevance induced by the compression branch. In particular, the normalized block-level attention score for query i, block m, and head h is

$$p_{i,m}^{\mathrm{cmp},h} = \frac{\exp\left(\left\langle \mathbf{q}_i^h, \bar{\mathbf{k}}_m^h \right\rangle / \sqrt{d_h}\right)}{\sum_{m'=1}^{M} \exp\left(\left\langle \mathbf{q}_i^h, \bar{\mathbf{k}}_{m'}^h \right\rangle / \sqrt{d_h}\right)}. \tag{11}$$

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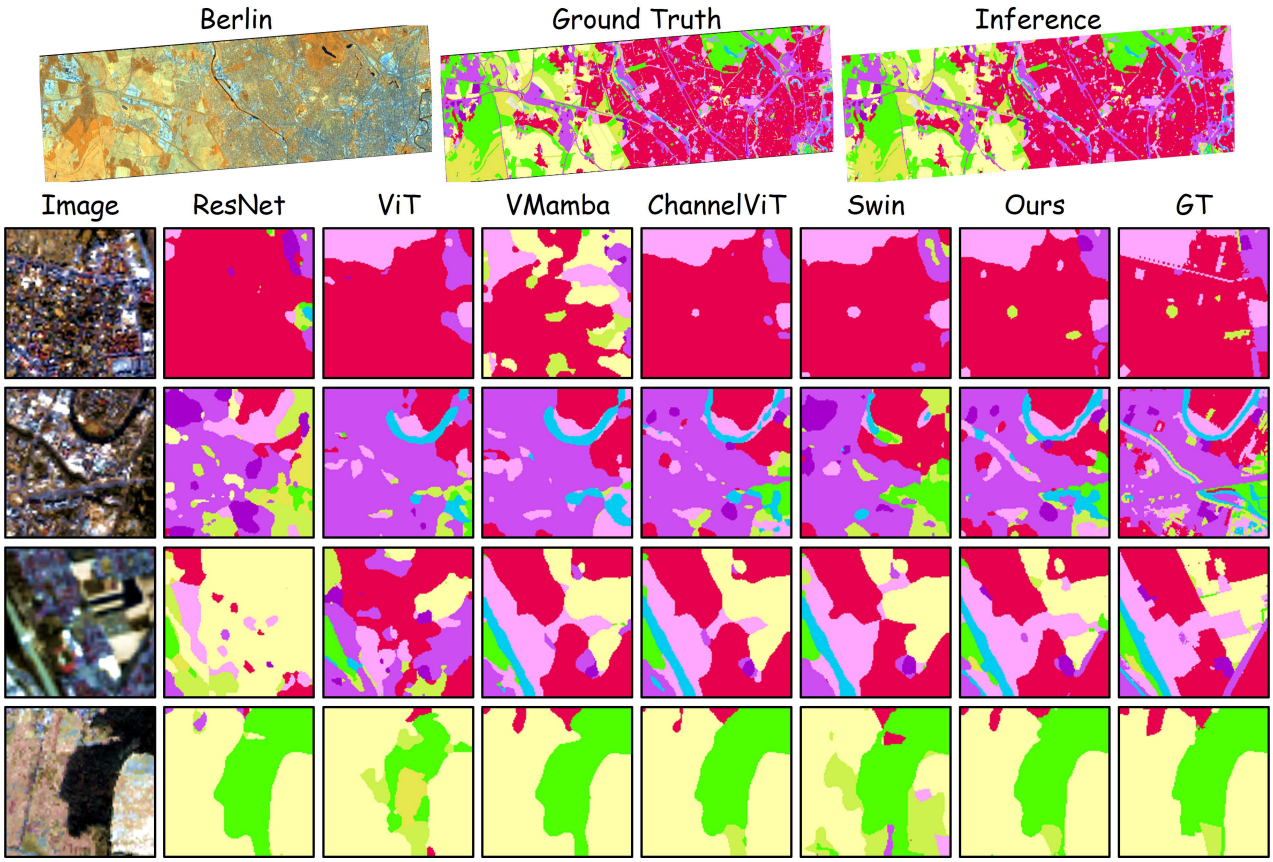


Fig. 6. Qualitative comparison on the C2Seg Berlin region. SpectralNSA better matches fine structures and produces more coherent local predictions.

TABLE IV
QUANTITATIVE COMPARISON ON **H2Crop**

Method	P	R	F1	OA	mIoU
			Tiny		
ResNet	44.47	50.90	46.81	76.90	35.82
ViT	46.24	49.87	47.48	77.62	36.37
VMamba	46.66	51.11	48.35	77.95	37.06
ChannelViT	45.54	52.89	47.83	78.59	37.20
Swin	44.69	**54.80**	47.63	77.48	36.81
SpectralNSA	**49.28**	50.46	**49.35**	**79.83**	**38.33**
			Small		
ResNet	43.82	49.98	45.73	75.73	34.71
ViT	47.89	49.34	48.02	77.91	36.74
VMamba	**47.94**	50.27	48.63	78.24	37.29
ChannelViT	46.56	53.00	48.78	78.91	37.93
Swin	47.04	52.68	48.30	79.20	37.58
SpectralNSA	47.38	**54.85**	**49.90**	**79.62**	**38.84**
			Base		
ResNet	42.19	48.09	43.96	73.74	32.92
ViT	**51.32**	48.43	48.43	78.18	37.02
VMamba	49.01	50.49	49.25	78.80	37.83
ChannelViT	48.30	52.56	49.89	80.35	38.98
Swin	47.19	51.26	48.77	79.76	38.04
SpectralNSA	50.34	**53.31**	**51.36**	**80.92**	**40.14**

If each attention head selects blocks independently, the resulting visibility pattern becomes highly fragmented. To avoid this, we divide the query heads into G routing groups and share the block-level routing result within each group. Let $\mathcal{H}_g$ denote the set of heads in group g. The group-level block score is defined as

$$\bar{p}_{i,m}^{(g)} = \frac{1}{|\mathcal{H}_g|} \sum_{h \in \mathcal{H}_g} p_{i,m}^{\mathrm{cmp},h}. \tag{12}$$

For the ith query, we retain only the top-K_s spectral blocks

$$\mathcal{I}_i^{(g)} = \mathrm{TopK}\left(\bar{p}_{i,:}^{(g)}, K_s\right). \tag{13}$$

The visible key/value sets of the selection branch are then given by

$$\begin{aligned} \mathbf{K}_i^{\mathrm{slc},h} &= \mathrm{Cat}_{m \in \mathcal{I}_i^{(g)}}\left(\bar{\mathbf{k}}_m^h\right) \\ \mathbf{V}_i^{\mathrm{slc},h} &= \mathrm{Cat}_{m \in \mathcal{I}_i^{(g)}}\left(\bar{\mathbf{v}}_m^h\right), \quad h \in \mathcal{H}_g. \end{aligned} \tag{14}$$

The branch output becomes

$$\mathbf{o}_i^{\mathrm{slc},h} = \mathrm{Attn}\left(\mathbf{q}_i^h, \mathbf{K}_i^{\mathrm{slc},h}, \mathbf{V}_i^{\mathrm{slc},h}\right). \tag{15}$$

The selection branch refines compression by attending only to selected contiguous block prototypes, preserving structured trainable sparsity.

The hard TopK operation and its discrete block indices are nondifferentiable, and we do not use a straight-through estimator or continuous relaxation. Gradients propagate through the queries, selected key/value prototypes, active attention computation, output projection, and fusion gate. Unselected blocks receive no selection-branch gradient for that query. Because

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Fig. 7. Qualitative comparison on H2Crop. Each row shows an input crop image, predictions from ResNet, ViT, VMamba, ChannelViT, Swin, and SpectralNSA, and the ground-truth crop-type mask. Red boxes highlight representative areas where SpectralNSA better captures fine-grained crop boundaries and small field parcels.

the scoring representations are shared with the differentiable compression branch, the underlying projections and block prototypes remain jointly optimized.

4) Local Spectral Window: Compression and selection provide global scanning and informative-block focusing, but may oversparsify local spectral continuity. We therefore introduce a window branch to preserve the neighborhood around each spectral token. Let W_s denote the window size. The local neighborhood of the ith spectral token is defined as

$$\mathcal{N}_{W_s}(i) = \left\{ j \,\middle|\, 1 \le j \le C,\ |j-i| \le \left\lfloor \frac{W_s - 1}{2} \right\rfloor \right\}. \tag{16}$$

The output of the window branch is

$$\mathbf{o}_i^{\text{win},h} = \text{Attn}\left(\mathbf{q}_i^h, \mathbf{K}^h_{\mathcal{N}_{W_s}(i)}, \mathbf{V}^h_{\mathcal{N}_{W_s}(i)}\right). \tag{17}$$

This branch maintains neighboring-band continuity and provides a stable local reference for the global compression and selective focusing branches.

D. SSS Block

Fig. 3 illustrates the overall structure of the proposed SSS block. The block follows a spectral-first information flow: spectral dependencies are first reorganized at each spatial location, shared-spatial prototypes are then constructed from the updated channelwise features, and the resulting spatial context is finally injected back into all channels through a gated residual pathway.

1) Spectral-First Residual Update: After spectral modeling, cross-band dependencies at each spatial location have been explicitly reorganized. Let the input to the current block be

$$\mathbf{X} \in \mathbb{R}^{C\times N\times D}. \tag{18}$$

For each spatial location n, we independently apply SpectralNSA to obtain the spectral update

$$\tilde{\mathbf{X}}_{:,n,:} = \text{SNSA}_{\text{spec}}\left(\text{LN}\left(\mathbf{X}_{:,n,:}\right)\right). \tag{19}$$

We then introduce a tokenwise fusion gate

$$\lambda_{c,n} = \sigma\left(\mathbf{w}_f^\top \tilde{\mathbf{x}}_{c,n}\right) \tag{20}$$

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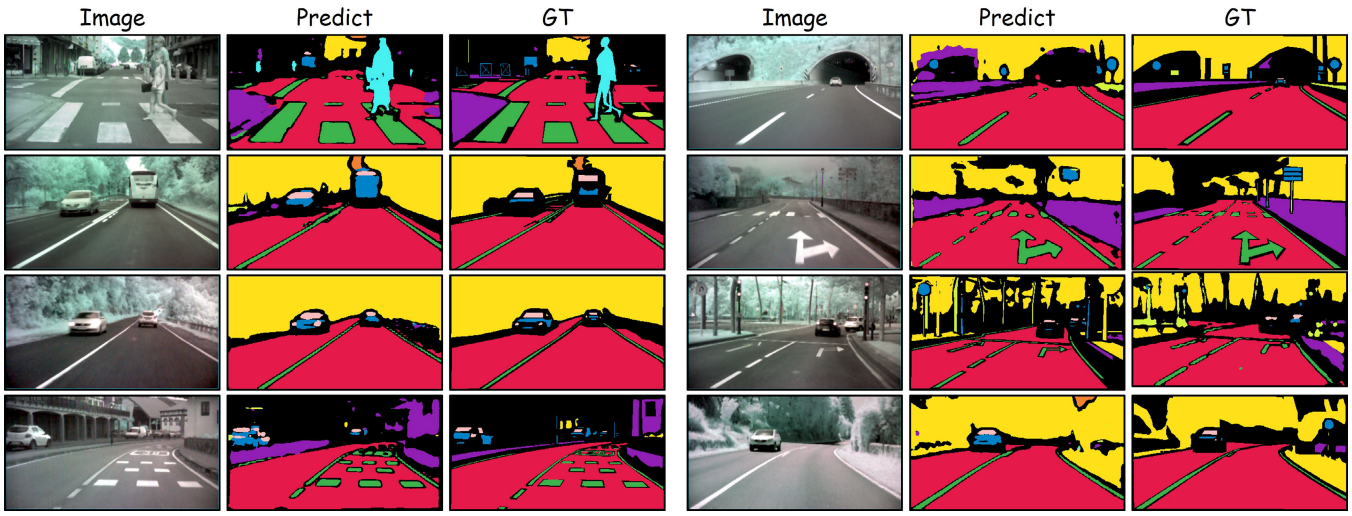


Fig. 8. Qualitative results on HSI-Drive. SpectralNSA produces material-aware segmentation maps that remain close to the ground truth across diverse driving scenes.

and perform a residual spectral update

$$\hat{\mathbf{x}}_{c,n} = \mathbf{x}_{c,n} + \lambda_{c,n}\tilde{\mathbf{x}}_{c,n}. \tag{21}$$

This step ensures that the subsequent spatial modeling is always built upon features that have already been spectrally screened and reorganized. In other words, spatial modeling is not performed in parallel with spectral modeling, but is introduced only after spectral interactions have been explicitly updated.

2) Shared-Spatial Prototype Construction: After the spectral update, applying spatial attention independently to each channel would be both computationally expensive and highly redundant, since many channels share similar spatial boundaries and region layouts. To address this, we first distill the multichannel features into a small set of **shared-spatial prototypes**, thereby constructing a more compact shared-spatial representation.

In particular, for each spatial location n, a learnable router predicts the channel weight for the kth shared-spatial prototype

$$a_{c,n,k} = \mathbf{w}_{r,k}^{\top}\hat{\mathbf{x}}_{c,n} \tag{22}$$

which is normalized across the channel dimension as

$$\pi_{c,n,k} = \frac{\exp\left(a_{c,n,k}\right)}{\sum_{c'=1}^{C}\exp\left(a_{c',n,k}\right)}. \tag{23}$$

The kth shared-spatial prototype is then defined as

$$\mathbf{u}_{k,n} = \sum_{c=1}^{C}\pi_{c,n,k}\hat{\mathbf{x}}_{c,n}, \quad k = 1,\ldots,K. \tag{24}$$

Collecting all prototypes yields $\mathbf{U} \in \mathbb{R}^{K\times N\times D}$.

3) Shared-Spatial Modeling: Once the shared-spatial prototypes have been obtained, spatial modeling need not be repeated across all C channels. Instead, we apply a blockwise sparse spatial attention operator to the compact prototype tensor $\mathbf{U} \in \mathbb{R}^{K\times N\times D}$

$$\tilde{\mathbf{U}} = \mathrm{SNSA}_{\mathrm{spa}}\left(\mathrm{LN}\left(\mathbf{U}\right)\right), \quad \tilde{\mathbf{U}} \in \mathbb{R}^{K\times N\times D}. \tag{25}$$

Here, sparse attention follows the compress–select–window principle, but the block partition is defined over the spatial token axis rather than the spectral channel axis. In implementation, the K prototype groups can be folded into the batch dimension or processed as a compact prototype set. This keeps spatial modeling low-redundant while maintaining a consistent sparse attention principle across spectral and spatial axes.

4) Prototype Aggregation and Gated Reinjection: After spatial modeling over the shared prototypes, the resulting spatial context must be injected back into the original channelwise representations. To this end, we first adaptively aggregate the K prototypes at each spatial location n

$$\beta_{k,n} = \frac{\exp\left(\mathbf{w}_s^{\top}\tilde{\mathbf{u}}_{k,n}\right)}{\sum_{k'=1}^{K}\exp\left(\mathbf{w}_s^{\top}\tilde{\mathbf{u}}_{k',n}\right)} \tag{26}$$

which produces the shared-spatial context

$$\mathbf{h}_n = \sum_{k=1}^{K}\beta_{k,n}\tilde{\mathbf{u}}_{k,n}. \tag{27}$$

We then broadcast $\mathbf{h}_n$ back to all channels and control its injection strength with a tokenwise gate

$$\rho_{c,n} = \sigma\left(\mathbf{w}_{\rho}^{\top}\hat{\mathbf{x}}_{c,n}\right) \tag{28}$$

$$\mathbf{y}_{c,n} = \hat{\mathbf{x}}_{c,n} + \alpha\,\rho_{c,n}\,\mathbf{h}_n \tag{29}$$

where α is a learnable scaling parameter. Stacking all $\mathbf{y}_{c,n}$ yields $\mathbf{Y} \in \mathbb{R}^{C\times N\times D}$. Finally, the block output is obtained through a feed-forward update

$$\mathbf{X}' = \mathbf{Y} + \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{Y}\right)\right). \tag{30}$$

Overall, the block follows a spectral-first and shared-spatial flow: spectral dependencies are reorganized first, compact spatial context is then extracted from shared prototypes, and this context is finally reinjected into channelwise representations.

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TABLE V
QUANTITATIVE COMPARISON ON **HSI–DRIVE**

Method	P	R	F1	OA	mIoU
Tiny					
ResNet	54.61	59.56	56.67	**83.14**	44.55
ViT	55.26	59.43	**58.06**	82.92	45.62
VMamba	54.69	60.22	56.92	82.41	44.52
ChannelViT	55.05	58.98	56.61	82.88	44.43
Swin	50.42	56.51	54.14	82.08	42.31
SpectralNSA	**55.60**	**61.46**	58.04	**83.14**	**45.78**
Small					
ResNet	54.75	58.99	56.55	82.63	44.48
ViT	**55.64**	61.24	57.97	82.98	45.75
VMamba	55.38	60.13	57.36	82.59	45.02
ChannelViT	53.26	59.53	57.31	82.74	44.98
Swin	53.62	56.33	54.58	82.10	42.66
SpectralNSA	55.40	**61.78**	**58.09**	**83.34**	**45.90**
Base					
ResNet	53.00	58.63	55.13	81.96	43.09
ViT	**57.25**	59.78	58.19	83.12	45.80
VMamba	56.26	59.98	57.70	82.95	45.28
ChannelViT	54.85	61.10	57.33	82.99	45.22
Swin	53.78	56.50	54.83	82.27	42.89
SpectralNSA	56.00	**61.41**	**58.33**	**83.46**	**46.09**

TABLE VI
QUANTITATIVE COMPARISON ON **HSI-Road**

Method	P	R	F1	OA	mIoU
Tiny					
ResNet	95.15	96.05	95.60	97.62	94.18
ViT	94.28	**96.57**	95.41	97.51	93.92
VMamba	94.69	95.80	95.25	97.43	93.73
ChannelViT	94.66	95.34	95.00	97.30	93.42
Swin	94.61	96.34	95.46	97.54	94.00
SpectralNSA	**95.17**	96.22	**95.69**	**97.67**	**94.30**
Small					
ResNet	94.95	96.31	95.62	97.63	94.26
ViT	**95.10**	95.85	95.48	97.55	94.02
VMamba	94.83	95.96	95.39	97.51	93.92
ChannelViT	94.64	96.25	95.44	97.53	93.97
Swin	94.74	96.22	95.48	97.55	94.02
SpectralNSA	95.09	**96.45**	**95.76**	**97.70**	**94.39**
Base					
ResNet	94.84	**96.58**	95.70	97.67	94.31
ViT	95.34	95.78	95.56	97.61	94.14
VMamba	94.87	96.02	95.44	97.54	93.98
ChannelViT	95.02	96.44	95.72	97.68	94.34
Swin	95.11	96.24	95.67	97.66	94.27
SpectralNSA	**95.39**	96.43	**95.91**	**97.79**	**94.60**

E. Multistage Hierarchical Backbone

Built upon the SSS block above, we construct a scalable multistage hierarchical backbone. Let the input of the sth stage be $\mathbf{X}^{(s)} \in \mathbb{R}^{C\times N_s\times D_s}$, where $N_s = H_s^{\mathrm{sp}} W_s^{\mathrm{sp}}$ is the number of spatial tokens at this stage and D_s is the feature dimension. Each stage consists of multiple blocks of the same type

$$\mathbf{X}^{(s,\ell+1)} = \mathcal{F}^{(s,\ell)}\left(\mathbf{X}^{(s,\ell)}\right), \quad \ell = 1,\ldots,L_s \tag{31}$$

where $\mathcal{F}^{(s,\ell)}$ denotes the ℓth SSS block in stage s.

When transitioning to the next stage, neighboring 2×2 spatial regions within each channel are merged, followed by a linear projection to increase the feature dimension. Denoting this downsampling operator by $\mathcal{M}(\cdot)$, we obtain

$$\mathbf{X}^{(s+1)} = \mathcal{M}\left(\mathbf{X}^{(s,L_s)}\right). \tag{32}$$

Since this downsampling changes only the spatial resolution and feature scale while preserving the underlying channelwise token organization, the same spectral-native sparse modeling and shared-spatial modeling mechanism can be reused in all subsequent stages. Across model scales, SpectralNSA adopts the same two-stage hierarchical design, while the stage widths, depths, attention heads, and sparse attention budgets are adjusted according to model capacity. The complete configurations are summarized in Table I.

Table I summarizes the complete stagewise configuration. The paired entries in "Embed dim," "Depths," and "Heads" correspond to Stages 1 and 2, respectively. Within each model scale, B_s, K_s, W_s, B_p, K_p, W_p, and K_{proto} are shared by the two stages, whereas the routing-group count is specified separately as G_1/G_2 to match the stagewise head counts.

F. Blockwise Sparse Attention Organization and Memory-Aware Execution

SpectralNSA uses regular blockwise sparse computation with memory-aware execution. Block, window, and routing settings define the sparse model, whereas chunking only changes the execution schedule. Contiguous partitioning provides a shared sparse coordinate system, group-shared routing regularizes access across heads, and chunking controls activation memory.

1) Regular Blockwise Sparse Layout: Consider an input sequence along the spectral axis

$$\mathbf{Z} = [\mathbf{z}_1,\ldots,\mathbf{z}_L] \in \mathbb{R}^{L\times D}. \tag{33}$$

To obtain a regular block layout, the sequence length L is padded to a multiple of the block size B

$$\hat{L} = \left\lceil \frac{L}{B} \right\rceil B \tag{34}$$

and the padded sequence is divided into $M = \hat{L}/B$ contiguous spectral blocks

$$\mathcal{B}_m = \{(m-1)\,B + 1,\ldots,mB\}, \quad m = 1,\ldots,M. \tag{35}$$

Padded positions are masked and do not participate in attention. Compression, selection, and local-window attention, therefore, operate in the same block coordinate system, allowing sparse visible regions to be represented by block indices rather than irregular token-level indices. Heads within each routing group share selected block indices, preserving adaptive selection while reducing fragmented per-head access.

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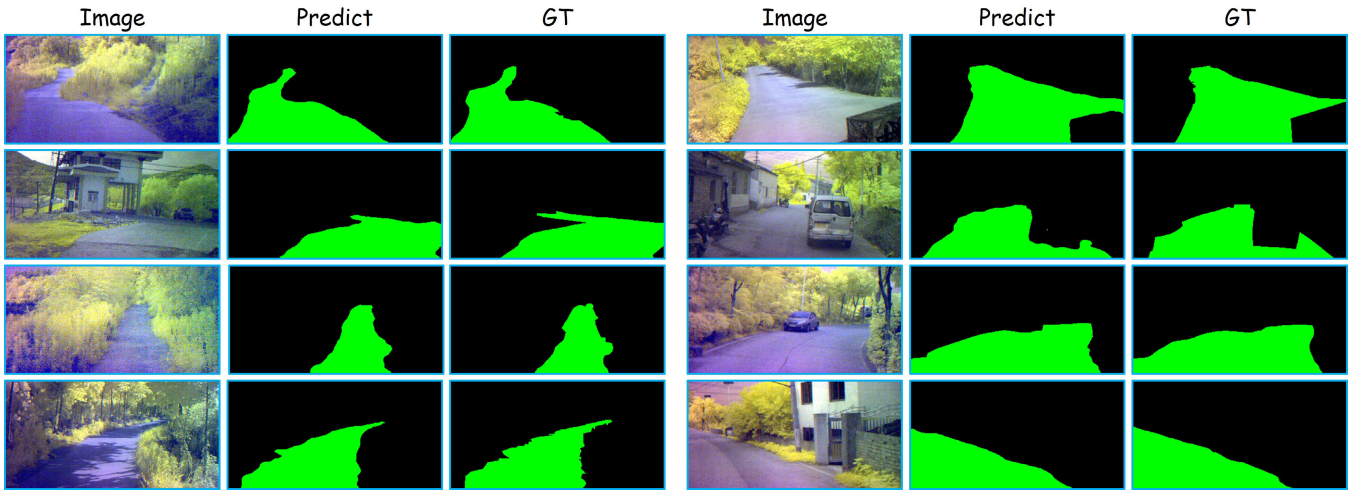


Fig. 9. Qualitative results on HSI-Road. SpectralNSA recovers road regions across urban and rural scenes while maintaining clean binary boundaries.

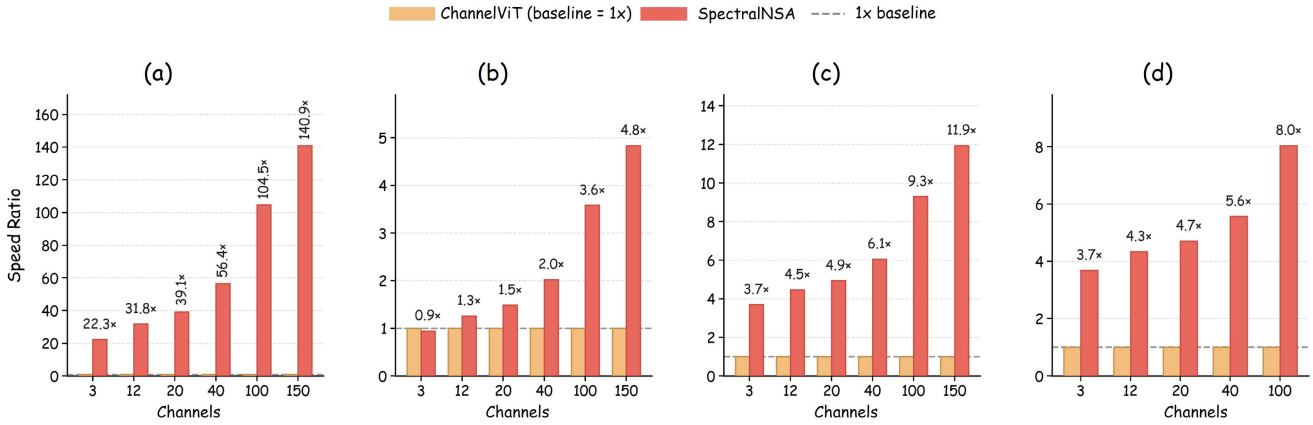


Fig. 10. FLOPs saving factor with increasing spectral channels. The spatial image size is fixed, and only the input channel number is varied. ChannelViT is normalized as the 1× baseline, while SpectralNSA bars report the FLOPs saving factor computed as ChannelViT/SpectralNSA. Larger values indicate stronger computational reduction by SpectralNSA. (a) Tiny. (b) Small. (c) Base. (d) Large.

2) Memory-Aware Spectral Chunked Execution: SpectralNSA models spectral dependencies independently at each spatial location. Therefore, the spatial dimension can be partitioned during execution without changing the spectral attention formulation (see Fig. 4). Let the full set of spatial locations be

$$\Omega = \{1, \ldots, N\}. \tag{36}$$

To control the size of intermediate activations at high spatial resolutions, Ω is divided into several nonoverlapping subsets

$$\Omega = \Omega_1 \cup \Omega_2 \cup \ldots \cup \Omega_T, \qquad \Omega_t \cap \Omega_{t'} = \varnothing \quad (t \neq t'). \tag{37}$$

The same spectral-NSA operator is then applied to each spatial chunk

$$\tilde{\mathbf{X}}_{:,\Omega_t,:} = \mathrm{SNSA}_{\mathrm{spec}}\left(\mathrm{LN}\left(\mathbf{X}_{:,\Omega_t,:}\right)\right), \quad t = 1, \ldots, T. \tag{38}$$

This changes only the execution order and preserves the per-location spectral attention result while reducing intermediate activation size. We use $N_c = 64$ by default in downstream experiments; standalone latency and sensitivity profiling use full spectral execution to avoid chunk-loop overhead.

TABLE VII

REPRESENTATIVE SPECTRAL–SPATIAL BACKBONE COMPARISON. HYPERSIGMA-B USES EXTERNAL PRETRAINING; THE OTHER ROWS ARE TRAINED FROM SCRATCH UNDER THE COMMON DOWNSTREAM PROTOCOL. PARAMETERS AND FLOPS ARE BACKBONE ONLY; ACCURACY IS MIOU (%)

Method	Pretrain	Params (M)	FLOPs (G)	H2Crop	HSI-Drive
SpectralFormer	No	5.78	7.92	39.18	45.34
SSFTT	No	1.84	4.36	38.67	45.08
HiT	No	9.47	12.65	39.41	45.49
SST	No	13.26	21.83	39.63	45.71
HyperSIGMA-B	Yes	174.62	112.73	**40.36**	**46.27**
ChannelViT	No	85.21	657.57	38.98	45.22
SpectralNSA	No	72.29	89.44	40.14	46.09

IV. EXPERIMENTS

A. Experimental Setup

We evaluate SpectralNSA on scene classification, semantic segmentation, hyperspectral driving perception, and pretraining transfer. Representative baselines cover CNN, transformer,

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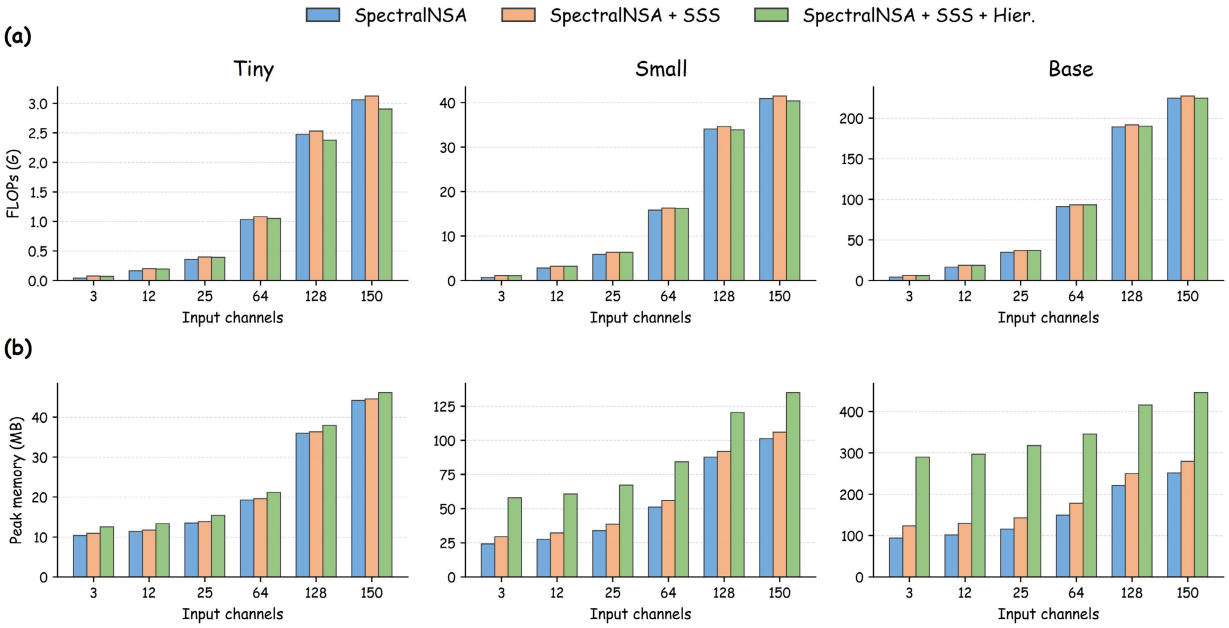


Fig. 11. Efficiency ablation of SpectralNSA internal components. FLOPs and inference memory are compared under tiny, small, and base settings as the number of input channels increases. "SpectralNSA" denotes the spectral sparse attention backbone only, "SpectralNSA+SSS" further adds the SSS block, and "SpectralNSA+SSS+Hier." denotes the full model with the multistage hierarchical backbone. The comparison isolates backbone components and excludes the task decoder. (a) Computational cost. (b) Memory footprint.

TABLE VIII

ABLATION OF THE SSS DESIGN. EUROSAT IS REPORTED WITH TOP-1 ACCURACY (%), WHILE C2SEG AND HSI-DRIVE ARE REPORTED WITH MIOU (%). THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

SSS Block	Multi-Stage	EuroSAT	C2Seg	HSI-Drive
		96.07	52.18	44.72
✓		<u>96.46</u>	<u>54.91</u>	<u>45.57</u>
	✓	96.14	52.67	44.88
✓	✓	**97.04**	**56.42**	**46.09**

TABLE IX

ABLATION OF THE NUMBER OF SHARED-SPATIAL PROTOTYPES K. EUROSAT IS REPORTED WITH TOP-1 ACCURACY (%), WHILE C2SEG AND HSI-DRIVE ARE REPORTED WITH MIOU (%)

K	EuroSAT	C2Seg	HSI-Drive
1	96.46	53.23	45.14
2	96.78	54.69	45.63
4	97.04	56.42	46.09
8	**97.21**	**56.80**	**46.67**

TABLE X

CONTROLLED COMPARISON OF NSA TOKEN ORGANIZATIONS. THE FIRST FOUR VARIANTS USE IDENTICAL SETTINGS AND DISABLE SHARED-SPATIAL MODELING; WAVELENGTH-SHUFFLED PRESERVES SAME-LOCATION GROUPING WHILE RANDOMIZING SPECTRAL ORDER

Variant	Shared spatial	H2Crop	HSI-Drive
Flattened-axis NSA	No	38.84	44.91
Random-grouped NSA	No	39.02	45.00
Wavelength-shuffled spectral-axis NSA	No	39.31	45.21
Spectral-axis NSA	No	<u>39.52</u>	<u>45.43</u>
Full SpectralNSA	Yes	**40.14**	**46.09**

TABLE XI

ABLATION OF THE THREE SPECTRAL ATTENTION BRANCHES. CMP., SEL., AND WIN. DENOTE COMPRESSION, TOP-k SELECTION, AND LOCAL-WINDOW BRANCHES, RESPECTIVELY

Variant	Cmp.	Sel.	Win.	H2Crop	HSI-Drive
w/o compression	–	✓	✓	39.58	45.62
w/o Top-k selection	✓	–	✓	39.29	45.43
w/o local window	✓	✓	–	<u>39.66</u>	<u>45.70</u>
Full SpectralNSA	✓	✓	✓	**40.14**	**46.09**

state-space, and channelwise spectral backbones: ResNet [38], ViT [39], VMamba [40], ChannelViT [7], and Swin transformer [13]. Tiny/small/base denote representative family specific capacity tiers rather than strict iso-parameter or iso-FLOPs matches; for ResNet, they correspond to ResNet-18/50/101. Scene classification uses top-1 accuracy, while dense prediction reports precision, recall, mean $F1$, overall accuracy (OA), and mIoU, with mIoU used as the primary ranking metric. Best and second-best results within each scale are highlighted in bold and underlined, respectively.

The complete two-stage configurations of SpectralNSA at tiny, small, base, and large scales are reported in Table I.

For dense prediction, all backbones use the same input bands, preprocessing, data splits, UPerNet decoder, cross-entropy loss, augmentation pipeline, optimizer components, schedule, and evaluation protocol. Each backbone selects its learning rate from the same predefined set $\{10^{-3}, 5 \times 10^{-4}, 10^{-4}, 5 \times 10^{-5}, 10^{-5}\}$ according to validation

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TABLE XII

ABLATION OF BRANCH FUSION AND ROUTING. "AVERAGE" USES EQUAL BRANCH WEIGHTS, "DYNAMIC GATE" USES TOKENWISE SOFTMAX FUSION, AND PER-HEAD/GROUP-SHARED DENOTE THE ROUTING GRANULARITY

Variant	Branch Fusion	Routing	H2Crop	Δ	HSI-Drive	Δ
Full branches + average fusion	Average	Group-shared	39.78	−0.36	45.75	−0.34
Full branches + per-head routing	Dynamic gate	Per-head	39.92	−0.22	45.88	−0.21
Full SpectralNSA	Dynamic gate	Group-shared	**40.14**	–	**46.09**	–

TABLE XIII

LEARNED VERSUS FIXED-RANDOM TOP-k ROUTING UNDER THE SAME SPARSE BUDGET ($K_s = 2$). ALL OTHER SETTINGS ARE UNCHANGED

Routing	H2Crop	HSI-Drive
Fixed-random Top-k	39.53	45.58
Learned Top-k	**40.14**	**46.09**

performance, so no method receives a broader search space. C2Seg experiments use only optical hyperspectral data and exclude SAR from the wavelength-ordered sequence. Sentinel-2-based datasets use the released spatially aligned products without additional bandwise resampling.

B. Quantitative Comparison With Representative Backbones

1) Multispectral Scene Classification on EuroSAT: **EuroSAT** is a Sentinel-2 land-use/land-cover benchmark with 27 000 image patches from 10 classes and all 13 spectral bands [41]. Table II presents that SpectralNSA achieves the best top-1 accuracy at tiny, small, and base scales. The consistent margin over ChannelViT is informative because both preserve channelwise tokens, while SpectralNSA additionally reorganizes backbone attention around spectral structure.

2) Semantic Segmentation on Mixed C2Seg Berlin–Augsburg Regions: **C2Seg** contains geographically distinct Berlin and Augsburg urban regions [42]. We use only the optical hyperspectral data and adopt a mixed-region protocol: Berlin and Augsburg samples are merged and split 4:1 into training and test subsets. Under this mixed-region protocol, SpectralNSA obtains the strongest overall results across all scales, including the primary mIoU metric (see Table III). Figs. 5 and 6 further show coherent land-cover regions and improved preservation of local structures.

We additionally evaluate within-sensor variable-band robustness without retraining by physically removing unavailable bands. C2Seg mIoU decreases from 56.42% with 150 bands to 55.36%, 54.12%, and 46.18% with 80, 50, and 10 bands, respectively. We also compare spectral descriptors without changing the tokenizer or backbone: a center-wavelength descriptor yields 40.43%/45.74% mIoU on H2Crop/HSI-Drive, compared with 40.14%/46.09% for canonical channel-index embeddings. The two descriptors exhibit comparable but dataset-dependent behavior.

3) Fine-Grained Agricultural Segmentation on H2Crop: **H2Crop** combines EnMAP hyperspectral imagery with Sentinel-2 time series for fine-grained agricultural segmentation [43]. The task emphasizes subtle interclass spectral differences and fine field boundaries. SpectralNSA achieves the best mIoU, OA, and $F1$ across all scales (see Table IV), while Fig. 7 shows more complete field regions and better preserved crop boundaries.

4) Driving-Scene Segmentation on HSI-Drive: **HSI-Drive** is a 25-band VNIR driving-scene dataset with 10 material-oriented classes collected under diverse seasons, times, weather, and road conditions [44]. SpectralNSA achieves the best mIoU at every scale and the best OA in the small and base settings (see Table V). Its recall/$F1$ profile and the qualitative examples in Fig. 8 indicate stable use of spectral cues under limited supervision and environmental variation.

5) Road Segmentation on HSI-Road: **HSI-Road** contains 3799 urban and rural scenes with 25 NIR bands and binary road masks over diverse surface materials [45]. Because strong baselines already achieve high scores, it provides a near-saturated test of consistent improvement. SpectralNSA nevertheless obtains the best mIoU at all scales and the strongest OA/$F1$ profile overall (see Table VI); Fig. 9 shows corresponding qualitative road masks.

Three-seed repetitions using seeds 42, 43, and 44 preserve the same mean ranking. SpectralNSA obtains 46.06% ± 0.08% mIoU on HSI-Drive and 94.58% ± 0.05% on HSI-Road, confirming stable but moderate margins.

6) Comparison With Representative Spectral–Spatial Backbones: To complement the generic-backbone comparisons above, Table VII further positions SpectralNSA against representative spectral–spatial architectures under a common dense-prediction protocol. The controlled from-scratch baselines use the same input channels, UPerNet decoder, preprocessing, loss, optimization schedule, learning-rate search space, and evaluation procedure. HyperSIGMA-B is reported separately with its publicly released external pretraining.

Among the methods trained from scratch, SpectralNSA achieves the highest mIoU on both H2Crop and HSI-Drive. HyperSIGMA-B is slightly higher by 0.22 and 0.18 points, respectively, but uses external pretraining and a substantially larger backbone (174.62- versus 72.29-M parameters). These results show that SpectralNSA remains competitive with dedicated spectral–spatial architectures.

C. Ablation Studies

1) Effect of the SSS Design: Table VIII illustrates the comparison of the spectral sparse backbone, SSS, the multistage hierarchy, and the full model. SSS provides a larger improvement than the hierarchy alone, indicating that compact spatial context is important after spectral modeling; combining both components gives the best results on all three datasets.

2) Effect of the Number of Shared-Spatial Prototypes: Table IX presents that increasing the number of shared-spatial prototypes consistently improves performance, while the gain from $K = 4$ to $K = 8$ is smaller than the gain from $K = 1$ to $K = 4$. We therefore use $K_{\text{proto}} = 2$ for tiny/small and $K_{\text{proto}} = 4$ for base to balance accuracy and efficiency.

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TABLE XIV

EFFECT OF SPECTRAL CHUNK SIZE N_c. MEMORY AND LATENCY ARE MEASURED FOR BACKBONE-ONLY SINGLE-SAMPLE AMP INFERENCE; MIOU IS EVALUATED WITH THE COMPLETE SEGMENTATION MODEL

Chunk size N_c	H2Crop			HSI-Drive		
	Peak mem. (GB)	Latency (ms/img)	mIoU	Peak mem. (GB)	Latency (ms/img)	mIoU
Full	8.10	58.2	40.14	5.40	41.8	46.09
128	6.35	64.7	40.14	4.46	47.2	46.09
64	5.20	72.8	40.14	3.84	54.5	46.09
32	4.45	91.6	40.13	3.38	68.9	46.08
16	4.10	122.4	40.13	3.15	93.2	46.08

TABLE XV

EFFICIENT/SPARSE TRANSFORMER COMPARISON UNDER A COMMON PROTOCOL. ACCURACY IS REPORTED AS MIOU (%)

Method	H2Crop	HSI-Drive
Performer-ViT	38.36	44.76
Longformer-ViT	38.79	45.07
BigBird-ViT	39.03	45.29
NSA-ViT	39.44	45.64
MoBA-ViT	39.28	45.51
VSA-ViT	39.56	45.77
ChannelViT	38.98	45.22
SpectralNSA	**40.14**	**46.09**

3) Effect of Spectral Sequence Organization: Table X separates sequence length, same-location grouping, and wavelength ordering. Random-grouped NSA is length-matched to spectral-axis NSA but removes same-location correspondence and spectral order; wavelength-shuffled NSA is stricter because it preserves the same C bands at each spatial location while randomizing their order. Spectral-axis NSA outperforms flattened-axis and random-grouped controls and further exceeds wavelength-shuffled NSA by 0.21/0.22 mIoU on H2Crop/HSI-Drive. This indicates that the gain is not explained only by a shorter sequence and that preserving wavelength order provides an additional benefit.

4) Three-Branch Sparse Attention: Removing any branch decreases mIoU on both datasets (see Table XI), confirming the complementary roles of global compression, selective block routing, and local spectral continuity.

5) Dynamic Fusion and Group-Shared Routing: Table XII presents that dynamic tokenwise fusion outperforms equal averaging, and group-shared routing provides a better accuracy–regularity tradeoff than independent per-head routing.

6) Learned Top-k Routing: Under the same sparse budget, learned Top-k routing outperforms fixed-random selection (see Table XIII), demonstrating the benefit of data-dependent routing.

7) Spectral Chunk Size: Table XIV presents that smaller chunks reduce peak memory at the cost of latency while leaving accuracy essentially unchanged; residual 0.01-point differences arise from AMP precision and rounding.

D. Efficiency and Scalability Analysis

We compare SpectralNSA with ChannelViT because both preserve channelwise patch tokens and, therefore, provide a closely matched view of how backbone computation changes after channelwise tokenization. We report FLOPs, single-sample inference latency, and inference memory under tiny/small/base/large settings and six channel numbers (3, 12, 20, 40, 100, and 150). All measurements use the same RTX 4090, single-sample AMP inference, and backbone-only forward pass. FLOPs saving factors are computed as ChannelViT/SpectralNSA. The maximum reduction of 140.94× is observed for tiny at 150 channels, while the ratio varies across model scales and channel counts.

1) Comparison With Efficient and Sparse Attention Baselines: Beyond the dense ChannelViT reference, we also compare with controlled efficient/sparse transformer variants in Table XV. Performer-, Longformer-, BigBird-, NSA-, MoBA-, and VSA-style attention are instantiated within the same channelwise ViT framework and evaluated with the same downstream protocol to compare alternative efficient-attention organizations.

SpectralNSA achieves the highest mIoU among the compared methods on both H2Crop and HSI-Drive (see Table XV), outperforming the strongest baseline, VSA-ViT, by 0.58 and 0.32 percentage points, respectively. These results demonstrate the benefit of organizing sparse interactions around the same location for semantic segmentation.

Figs. 10–12 and Table XVI show that SpectralNSA becomes increasingly favorable in FLOPs and memory as the number of spectral channels grows. The Small three-channel setting is an exception in FLOPs because fixed routing, prototype, and tensor-layout overheads dominate at very short spectral sequences. Latency is also less monotonic at low channel counts for the same reason, but the trend reverses as dense channelwise interaction becomes the main bottleneck.

At the largest setting, ChannelViT is not reported for large with 150 channels because it exceeds available memory, whereas SpectralNSA remains executable. FLOPs and measured latency need not follow exactly the same trend because latency also depends on memory access, kernel launches, sparse routing, tensor transformations, attention backends, and GPU utilization. Overall, SpectralNSA exhibits its strongest scalability advantage for moderate- and high-dimensional spectral inputs.

2) Computational and Memory Effects of Internal Modules: Fig. 11 and Table XVII separate the spectral sparse backbone, SSS block, and hierarchy. SSS adds modest FLOPs because

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TABLE XVI

FLOPS COMPARISON BETWEEN CHANNELVIT AND SPECTRALNSA UNDER DIFFERENT MODEL SCALES. RATIO IS COMPUTED AS CHANNELVIT/SPECTRALNSA; LARGER VALUES INDICATE GREATER FLOPS REDUCTION ACHIEVED BY SPECTRALNSA. "–" INDICATES THAT CHANNELVIT RUNS OUT OF MEMORY ON THE TEST DEVICE

Channels	Tiny			Small			Base			Large		
	Channel ViT	Spectral NSA	Ratio↑	Channel ViT	Spectral NSA	Ratio↑	Channel ViT	Spectral NSA	Ratio↑	Channel ViT	Spectral NSA	Ratio↑
3	4.148	**0.186**	**22.30**×	**11.931**	12.594	0.95×	68.557	**18.540**	**3.70**×	241.047	**65.315**	**3.69**×
12	22.096	**0.694**	**31.84**×	58.679	**46.592**	**1.26**×	306.585	**68.700**	**4.46**×	1049.569	**241.873**	**4.34**×
20	45.056	**1.152**	**39.11**×	114.244	**76.979**	**1.48**×	560.198	**113.413**	**4.94**×	1880.343	**399.147**	**4.71**×
40	131.302	**2.327**	**56.43**×	310.850	**153.624**	**2.02**×	1367.310	**225.701**	**6.06**×	4418.821	**793.687**	**5.57**×
100	637.295	**6.098**	**104.51**×	1395.175	**389.354**	**3.58**×	5272.170	**566.914**	**9.30**×	15990.323	**1988.894**	**8.04**×
150	1342.267	**9.524**	**140.94**×	2865.403	**592.436**	**4.84**×	10226.093	**856.237**	**11.94**×	–	2998.181	–

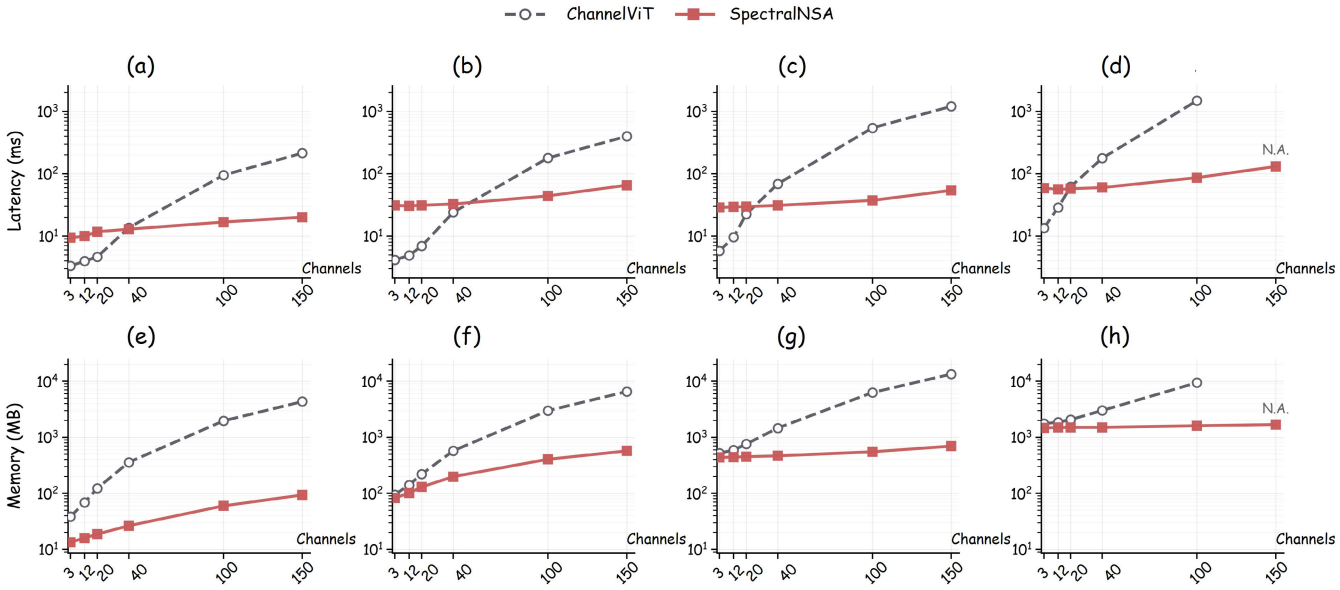


Fig. 12. Latency and memory comparison under increasing spectral channel numbers.The spatial image size is fixed, and the channel number is varied from 3 to 150. The top row reports single-sample inference latency, and the bottom row reports inference memory footprint under (a) and (e) tiny, (b) and (f) small, (c) and (g) base, and (d) and (h) large settings. Lower values are better. N.A. denotes settings where ChannelViT runs out of memory on the test device. Latency and memory are measured for backbone-only single-sample inference.

TABLE XVII

PARAMETER ABLATION OF SPECTRALNSA, THE SSS BLOCK, AND THE MULTISTAGE HIERARCHICAL BACKBONE. "SPEC." DENOTES SPECTRALNSA ONLY, "SPEC.+SSS" DENOTES SPECTRALNSA WITH THE SSS BLOCK USING $K = 4$, AND "FULL" DENOTES THE COMPLETE SPEC.+SSS+HIER. DESIGN

Scale	Spec.	Spec.+SSS	Full
Tiny	0.21–0.22	0.28–0.29	0.71–0.72
Small	3.58–3.61	4.79–4.82	12.19–12.22
Base	21.35–21.41	28.51–28.56	72.24–72.29

Note: Parameter counts are reported in millions (M). Each range corresponds to input channel numbers from 3 to 150.

spatial modeling operates on a small prototype set, whereas the hierarchy and intermediate prototype/gating features increase parameters and memory. Parameter counts remain nearly constant across channel numbers, indicating that the added capacity comes mainly from stage width and architectural modules rather than channel-specific parameter growth.

E. Pretraining and Transferability

We examine whether SpectralNSA can serve as a pretrainable backbone rather than only a task-specific architecture. Base-scale models are pretrained on **BigEarthNet** [46] using contrastive learning (CL) and MAE, and then fine-tuned on EuroSAT for scene classification, BigEarthNet for multilabel classification, and the 10-band SegMunich semantic-segmentation benchmark [2]. ViT, Swin, ChannelViT, and SpectralNSA use the same scratch, CL, and MAE protocols summarized in Table XVIII. For BigEarthNet downstream evaluation, validation and test images are excluded from self-supervised pretraining, model selection, and hyperparameter tuning. SpectralNSA achieves the best results under both CL and MAE, demonstrating its effectiveness as a pretrainable backbone across classification and segmentation tasks.

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TABLE XVIII

BACKBONE COMPARISON UNDER COMMON SCRATCH, CL, AND MAE PROTOCOLS. EUROSAT, BIGEARTHNET, AND SEGMUNICH ARE EVALUATED BY TOP-1 ACCURACY, MAP, AND MIOU, RESPECTIVELY

Pretraining	Backbone	EuroSAT	BigEarth Net	SegMunich
Scratch	ViT	94.09	76.92	45.31
	Swin	95.86	78.18	46.42
	ChannelViT	95.92	78.54	46.85
	SpectralNSA	**97.04**	**79.21**	**47.70**
CL	ViT	97.52	82.16	47.62
	Swin	97.91	82.94	48.24
	ChannelViT	98.12	83.41	48.76
	SpectralNSA	**98.46**	**84.58**	**49.30**
MAE	ViT	98.05	83.18	48.26
	Swin	98.38	83.88	48.93
	ChannelViT	98.57	84.46	49.44
	SpectralNSA	**98.93**	**85.37**	**50.60**

V. CONCLUSION

In this article, we revisit transformer backbone design after channelwise tokenization for optical RS spectral images. We propose **SpectralNSA**, a spectral-NSA backbone that models cross-band tokens at each spatial location through blockwise spectral compression, group-shared Top-k block selection, and local spectral windows. We further introduce a SSS structure that extracts compact shared-spatial context and reinjects it into channelwise representations, reducing redundant per-channel–spatial modeling. Combined with regular blockwise sparsity and memory-aware spectral chunking, SpectralNSA improves scalability while preserving structured spectral modeling. Experiments across classification, dense prediction, efficiency analysis, and pretraining transfer demonstrate a favorable accuracy–scalability tradeoff and competitive transferability. Remaining errors mainly involve spectrally similar classes, small or fragmented regions, and ambiguous object boundaries. These results indicate that optical RS Transformers should move beyond unified input interfaces toward backbone attention designed for long spectral sequences. Future work will explore cross-sensor wavelength harmonization, larger-scale pretraining, interpretable routing, and block-sparse operator optimization for improved generalization and deployment efficiency.

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Xuyang Li received the B.S. degree from the School of Optoelectronic Science and Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2023. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include machine learning, remote sensing foundation models, and multimodal interpretation.



Chenyu Li received the B.S., M.S., and Ph.D. degrees from Southeast University, Nanjing, China, in 2018, 2021, and 2025, respectively.

Her research interests include interpretable artificial intelligence, big Earth data forecasting, foundation models, and hyperspectral imaging.

Dr. Li received the outstanding Student Paper (Paul Gader) award at WHISPERS2025 and the Best Reviewer Award from IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS).



Lei Wang (Member, IEEE) received the B.Sc. and M.Sc. degrees from Shandong University, Jinan, China, in 1995 and 1998, respectively, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2001.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



Keming Chen (Member, IEEE) received the Ph.D. degree in pattern recognition and intelligent systems from the National Laboratory of Pattern Recognition (NLPR), Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2010.

Since 2011, he has been a Researcher with the Institute of Electronics, Chinese Academy of Sciences, where he is currently a Professor with the Aerospace Information Research Institute. His research interests include remote sensing image processing and machine learning.



Danfeng Hong (Senior Member, IEEE) received the Dr.-Ing. degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

He is currently a Full Professor at Southeast University, Nanjing, China, and previously served as a Full Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include artificial intelligence, multimodal perception, foundation models, hyperspectral remote sensing, and Earth science.

Dr. Hong serves as an Associate Editor for IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS). He has been recognized as a Highly Cited Researcher by Clarivate Analytics since 2022.

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